TEXORA · SESSIONAL NOTES · STUDY SHEET

Line Balancing and Apparel Costing

Balancing a sewing line from its operation bulletin — pitch time, operator allocation and efficiency — and building the garment's price from fabric, trims and cost of making, worked end to end.

AM-4202 · ADVANCED APPAREL MANUFACTURING-II SESSIONAL · STUDY 5 · LEVEL 4 TERM 2

1. Aim and place of the study

Aim: to balance a real sewing line from an operation bulletin and a production target, and to cost the same garment completely — the two calculations an apparel engineer performs more than any others, and the quantitative core of this sessional (the AM-4201/AM-3207 theory of IE and costing is applied here, not repeated).

2. Data used in this study

Garment: men's basic long-sleeve shirt, size range S–XL, to the TE-2207 chart (size M: chest 107, length 73, shoulder 46, sleeve 61.5, collar 40). **Operation bulletin** (SMV from time study at 100% rating, including machine and handling allowances) — given in Table 1. **Target:** 800 pieces per 8-hour shift (480 attended minutes). **Cost data:** §6.

3. The balancing method

Three quantities govern every line. The **pitch time** is the time the line may spend on each garment if it is to meet the target:

$$\text{pitch time} = \text{available time} \div \text{target output}$$

Each operation then needs, in theory, **SMV ÷ pitch** operators; an operation needing more time than the pitch is split across several operators working in parallel, each producing a part every pitch interval. The line is *balanced* when every station's workload per garment equals the pitch, so no station starves or floods its neighbour. The check on the whole arrangement is the **line efficiency**:

$$\text{efficiency} = (\text{SMV} \times \text{pieces produced}) \div (\text{operators attended} \times \text{minutes attended}) \times 100$$

4. Worked balancing: the shirt line

pitch = 480 min ÷ 800 pieces = 0.60 min per piece → line rate = 60 ÷ 0.60 = 100 pieces/hour

NO.	OPERATION	SMV (MIN)	OPERATORS = SMV ÷ 0.60
1	Collar making	1.80	3
2	Cuff making	1.20	2
3	Placket making and attaching	2.40	4
4	Pocket attaching	1.20	2
5	Yoke joining	0.60	1
6	Shoulder joining	1.80	3
7	Sleeve attaching	2.40	4
8	Side seaming	1.20	2
9	Collar attaching	1.80	3
10	Cuff attaching	1.20	2
11	Buttonholing	1.20	2
12	Button attaching	1.20	2
13	Bottom hemming	0.60	1
	Total SMV	**18.60**	**31 operators**

The independent check agrees: total operators = total SMV ÷ pitch = 18.60 ÷ 0.60 = **31**. Figure 1 shows why the division is necessary — operations 3 and 7 each contain four pitch-times of work, so four parallel operators each hand over a part every 0.60 min; operations 5 and 13 fit one operator exactly.

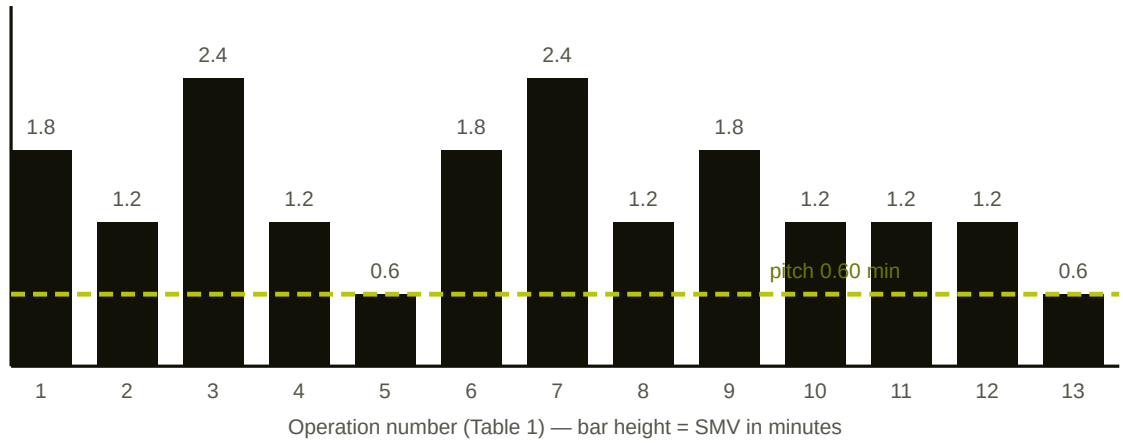


Figure 1 — Work content per operation against the pitch time. Any bar above the pitch line must be split across parallel operators; the number of operators is the bar height divided by the pitch.

5. Efficiency of the balanced line

The line as actually staffed carries **34 attended workers** (the 31 operators of Table 1 plus helpers and the line's share of inspection) and produces the target **800 pieces** in the 480-minute shift:

$$\text{efficiency} = (18.60 \times 800) \div (34 \times 480) \times 100 = 14{,}880 \div 16{,}320 \times 100 = 91.2\%$$

Read the figure correctly: the *balance* is perfect (every station loaded to the pitch), yet efficiency is 91.2%, because efficiency also counts the non-sewing attended minutes — helpers, waiting, bundle handling — in its denominator. The two numbers answer different questions, and confusing them is the commonest error in line reports. If output fell to 760 pieces with the same crew, efficiency would fall to (18.60 × 760) ÷ 16,320 = 86.6% with the balance unchanged.

6. Worked costing of the same shirt

Cost of making (CM) first. The line's daily cost: 34 workers × Tk 650 average daily wage = Tk 22,100; helpers, quality and supervisory share Tk 6,200; machine, power and factory-overhead allocation Tk 7,700 — a daily line cost of **Tk 36,000**. At 800 pieces per day:

$$\text{CM per piece} = 36{,}000 \div 800 = \text{Tk } 45.00$$

COST ELEMENT	BASIS	TK / PIECE
Fabric	1.50 m × Tk 240 per m	360.00
Trims and accessories	Buttons, thread, labels, tags, polybag, carton share	42.00
Cost of making (CM)	Line cost Tk 36,000 ÷ 800 pieces	45.00

Garment wash	Light wash, per piece	12.00
Production cost		**459.00**
Wastage allowance 2%		9.18
Cost after wastage		**468.18**
Margin 5% → **FOB price per piece**		**491.59**
FOB per dozen (× 12)		5,899.07

Three costing disciplines are on display in this small table. First, every element carries its **basis** — a price without a basis cannot be checked or renegotiated. Second, fabric dominates: Tk 360 of the Tk 459 production cost (78%) is fabric, which is why the marker efficiency of Study 4 matters more to the price than any saving in the sewing room. Third, allowances are **separate lines** (wastage 2%, margin 5%), never buried inside the elements — a buyer auditing the costing can see and challenge each one.

Precautions

- Balance from measured SMV, never from the supervisor's estimate: a 10% SMV error becomes a permanently mis-staffed line.
- Recount attended workers honestly in the efficiency denominator — excluding helpers flatters the figure and destroys its use for planning.
- State the currency, the date basis of prices, and the allowance percentages on every costing sheet; prices in this study are illustrative teaching values, not market quotations.

Practice (viva preparation)

Answer in your own words; brief model answers follow.

1. A garment has total SMV 15.0 min and the target is 600 pieces per 480-minute shift. Find the pitch time and the theoretical operator count.
2. The line of Question 1 runs with 21 attended workers and makes its 600 pieces. Find the efficiency, and explain why it is not 100% although the target was met.
3. A costing shows: fabric 1.90 m × Tk 210 = Tk 399; trims Tk 55; CM Tk 38; no wash. Build the price with 2% wastage and 5% margin.
4. In Table 1, why do operations 3 and 7 need four operators while operation 5 needs one?

Answer key

1. Pitch = 480 ÷ 600 = 0.80 min. Operators = 15.0 ÷ 0.80 = 18.75, so 19 operators in practice, with the small surplus absorbed as float or shared help.
2. Efficiency = (15.0 × 600) ÷ (21 × 480) × 100 = 9,000 ÷ 10,080 × 100 = 89.3%. The target was met, but the attended minutes include 21 workers while the work content corresponds to 18.75; the extra attended time lowers efficiency even at full output.
3. Production cost = 399 + 55 + 38 = Tk 492.00. Wastage 2% → 492 × 1.02 = Tk 501.84. Margin 5% → 501.84 × 1.05 = Tk 526.93 FOB per piece.
4. Each operator can deliver one part every pitch interval (0.60 min). Operations 3 and 7 contain 2.40 min of work = 4 pitch units, so four operators in parallel are needed to keep the line rate; operation 5 contains exactly one pitch unit (0.60 min), so one operator keeps pace.

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